

Lecture 3

Probability and probability distributions: Video 1 RECAP

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Probability as a **model** of the **data generating process**. Random processes have uncertainty; a probability model is meant to tell us the relative frequency of certain outcomes in the population.



Key concepts:

- Outcome
- Random process
- **Sample space**: sometimes denoted using set notation $\{ \}$, may consist of discrete or continuous outcomes
- Event
- **Probability** $P(\text{event})$ or $Pr(\text{event})$



Rules of probability

Axioms:

1. $P(A) \geq 0$ for some event A
2. $P(S) = 1$ for the sample space S
3. If $A_1, A_2, \dots, A_N$ are **mutually exclusive** events, then:
$$P(A_1 \cup A_2 \cup \dots \cup A_N) = P(A_1) + P(A_2) + \dots + P(A_N) = \sum_{i=1}^N P(A_i)$$

From this:

- $P(A) = 1 - P(\sim A)$
- For mutually exclusive events: $P(A \cup B) = P(A) + P(B)$
- Non-mutually exclusive events: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Conditional probability

Let $P(A|B)$ be the **conditional probability** that event A occurs given that event B occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(A \cap B) + P(\sim A \cap B)} = \frac{P(A \cap B)}{P(B)}$$

Bayes' Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$P(A)$ is a *prior*. $P(A|B)$ is a *posterior*.



Independence

- If A and B are **independent events**, the probability that A occurs does not depend on whether B occurred, and vice versa.
- Put another way, knowing whether B occurred provides *no information* about the probability that event A occurred (and vice versa).
- When A and B are independent events, $P(A|B) = P(A)$
- When A and B are independent events, $P(A \cap B) = P(A) \times P(B)$.
The *multiplicative rule*.
- The outcomes of sampling *with replacement* are independent events.

